Lab Procedure
Infrared Distance

Introduction

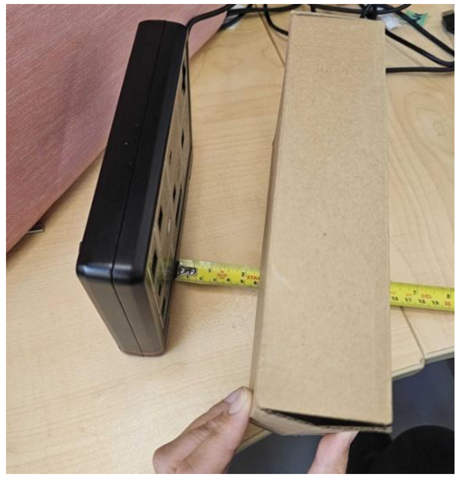
Ensure the following:

1. You have reviewed the Application Guide – Infrared Distance
2. Make sure the **Mechatronic Sensors Trainer** is out of its case; it is powered using the provided USB cable connected to the PC.
    a. The LED next to the USB C port will stay white after the touch screen starts loading.
    b. The load screen with the Quanser Logo should disappear after a few seconds and you should see some evenly spaced crosses on a black background on the touch screen.

Observing Infrared Distance Data

1. Launch Visual Studio Code or your preferred Python IDE. If using Visual Studio Code, click on **File** > **Open Folder…**, and browse to the directory containing the python files and lab procedure for the Infrared Distance lab (…/sensors_trainer/1_fundamentals/infrared_distance/hardware/python). Open this directory.
2. We will be recording data using the infrared Distance sensor. Open the datasheet in the same folder, **GP2Y0A21YK0F_Spec_Sheet.pdf** and identify the minimum and maximum measurement range.
3. Draw a distance ruler from the front of the Mechatronic Sensor Trainer, where the infrared distance sensor is, start from 0cm to the maximum range. You could also set up a measuring tape as shown below.

4. Grab a flat surface and place it so the flat surface is parallel to the front of the mechatronic sensor board as shown below.



5. Run infrared_display.py using the ▷ button or the terminal. Slowly move the flat surface close to the Mechatronics trainer and further away.

6. How does the voltage measurement look like?

Calibrating for Infrared Distance Sensor

7. Open infrared_record.py. This experiment will require you to record voltage measurements at different distances. These will then get used to generate a power trendline that follows an equation such as:

$$d = Ax^b \tag{1}$$

   Where x is the measured voltage output of the sensor and d is the distance in meters.

8. You will need to update some parameters on the experiment constants section:

   a. `minMeasurementValue` (the closest distance the sensor can read)

   b. `stepBetweenMeasurements` (how far apart are measurements you will take)

   c. `maximumMeasurementValue` (the farthest distance the sensor can read)

   Since we are using sampling methods to identify the value of A and b, it's important to generate as many samples as possible. We are also assuming the distance value being calculated is in meters.

9. Place a flat object at your marker for the minimum/closest distance. It's recommended to keep the flat object parallel to the front surface of the Mechatronic Sensors Trainer.

10. Run infrared_record.py using the ▷ button or the terminal. Click on the terminal session and use the spacebar on your computer to record a sample. It will show you what distance you should be sampling at based on your defined parameters.

11. Move the object based on the distance specified for `stepBetweenMeasurements`.

12. Once the total number of samples is reached a line fitting technique will be used to estimate A and b and a plot will show the predicted model of the infrared distance sensor compared to the actual sampled value.
Note: This does not have to be done in code directly, you could also note your distances and the output voltage in a software like Excel, plot the results and find the best fitting type of trendline.

13. Do the calibrated A and b values match the distance/voltage measurements? If so, make note of the A and b values. Take a screenshot of the output graph for the recommended assessment.

14. Does your graph look similar to the distance measuring graphs shown in the datasheet?

Testing Calibration

15. Open infrared_distance.py and in the experiment constants region, change:

    a. `amplitude` to the calculated value of A.

    b. `exponent` to the calculated value of b.

16. Run infrared_ distance.py using the ▷ button or the terminal.

17. A plot will show you both the Voltage and Distance measurements from the Infrared Distance sensor. Keep in mind the Distance is in meters.

18. Using a measuring tape or distance marks to verify the distances, move the infrared sensor to the distances used for calibration and make a note of the displayed distance values. Do they match correctly? What other considerations might you need to take when using an infrared distance sensor?

19. How well does the model you created in the previous section match the actual physical distance?

20. Stop, save and close your program.

21. Power OFF the Mechatronic Sensors Trainer by disconnecting its USB C cable, disconnect any external sensors and pack them along the base and the USB cables in the provided case.